

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0364

Woodmere 100 kV
February 15, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0364. The Interconnection Customer proposes a 329.3 MW wind facility interconnecting at the Woodmere 100 kV substation in the Ohio area. The request was received on August 18, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 195 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$45,871,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WOODMERE - MAGNOLIA RANCH NORTH SUBSTATION 100.0 kV line	WOODMERE - THUNDER 100.0 kV	241.9	128	100.0
WOODMERE - THUNDER 100.0 kV	WOODMERE - MAGNOLIA RANCH N 100.0 kV	225.2	137	100.0
MAGNOLIA RANCH NORTH - LICK 100.0 kV	SUN VALLEY - MAGNOLIA RANCH N 100.0 kV	132.1	129	33.9
LICK - SONORA 100.0 kV	MAGNOLIA RANCH N - SUN VALLEY 100.0 kV Line	131.3	130	33.9
THUNDER - MAGNOLIA RANCH N 100.0 kV	WOODMERE - MAGNOLIA RANCH NORTH 100.0 kV	120.3	143	28.7
SUN VY - MAGNOLIA RANCH N 100.0 kV L	MAGNOLIA RANCH N - LICK 100.0 kV Line	118.4	132	29.0
SUN VLY - NORSEAL 100.0 kV line	MAGNOLIA RANCH N - LICK 100.0 kV Line	113.4	129	29.0
PORTAGE LAKES - OLD FORGE 230.0 kV	SURROWIEC - WEEDPATCH 500.0 kV Line	107.0	833	5.9
SPORN - GAVIN POWER LLC SUBSTATION 230.0 kV	INTERNATIONAL PAPER - FLATLICK 500.0 kV	102.8	1,196	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WOODMERE - MAGNOLIA RANCH NORTH SUBSTATION 100.0 kV line	Rebuild line with higher-capacity conductor	\$9,205,000	100% (\$9,205,000)
WOODMERE - THUNDER 100.0 kV	Rebuild line with higher-capacity conductor	\$4,473,000	100% (\$4,473,000)
MAGNOLIA RANCH NORTH - LICK 100.0 kV	Rebuild line with higher-capacity conductor	\$2,441,000	100% (\$2,441,000)
LICK - SONORA 100.0 kV	Rebuild line with higher-capacity conductor	\$2,294,000	100% (\$2,294,000)
THUNDER - MAGNOLIA RANCH N 100.0 kV	Rebuild line with higher-capacity conductor	\$10,683,000	100% (\$10,683,000)
SUN VY - MAGNOLIA RANCH N 100.0 kV L	Rebuild line with higher-capacity conductor	\$2,968,000	100% (\$2,968,000)
SUN VALLEY - NORSEAL 100.0 kV	Rebuild line with higher-capacity conductor	\$12,731,000	100% (\$12,731,000)
PORTAGE LAKES - OLD FORGE 230.0 kV	Reconductor line	\$13,880,000	7% (\$1,002,000)
SPORN - GAVIN POWER LLC SUBSTATION 230.0 kV	Reconductor line	\$3,635,000	2% (\$74,000)
	Total allocated to project		\$45,871,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.